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Lab Report

INTRODUCTION TO ROBOTIC AND INTELLIGENT SYSTEM LAB

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Title: Arduino Photoresistor Control for Motor

Objective:

The project is to implement a sensor, photoresistor (LCD), to detect the intensity of daylight brightness. When the sensor identifies a sunny day, it will activate a motor to power a fan-driven cooling process.

Simulate Arduino by Tinkercad

Tinkercad allow us to simulate and design electronic circuits using Arduino boards and components.

1. employ a photoresistor for light-intensity detection

2. The circuit's resistance varies according to the level of light intensity (thereby influencing the resistance alteration in the circuit and affecting the motor).

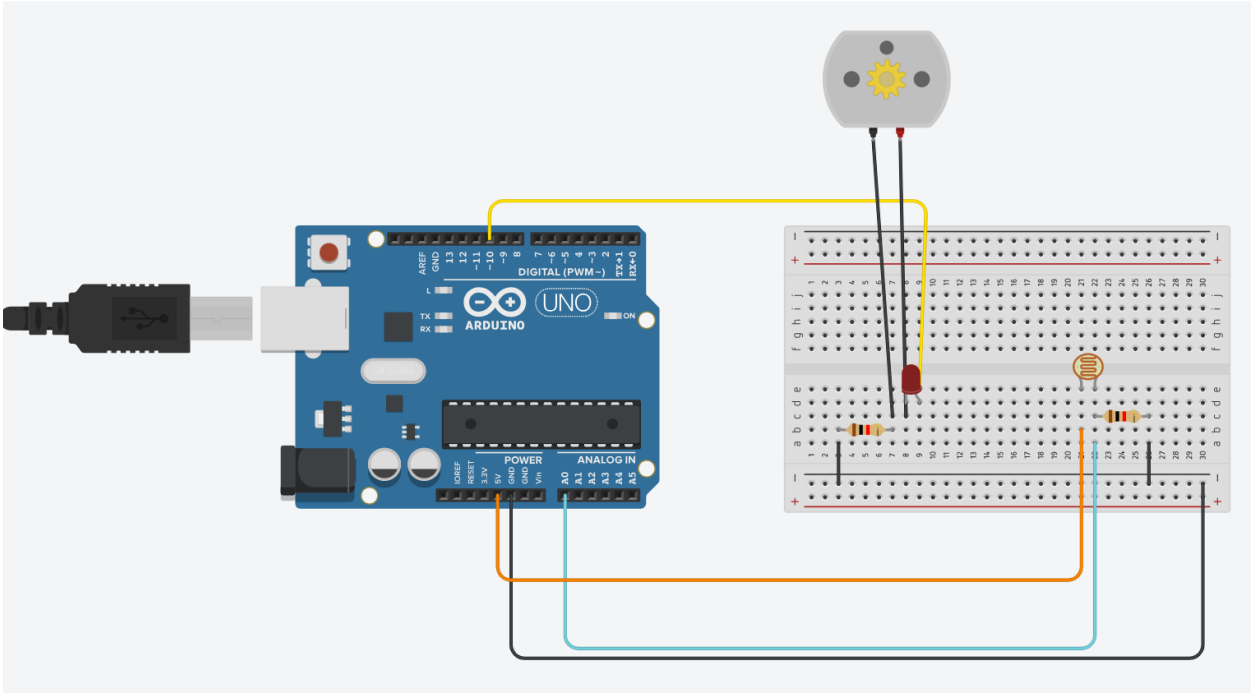
3. Use a LED to tell that the motor is working:

   Light intensity = high --> Power on

   Light intensity = low --> Power off

4. The motor, in turn, transforms electrical energy into mechanical energy.

Circuit Diagram



Attach the photoresistor to pin A0, and the outside pins to +5V and ground.

Reads an analog input (photoresistor) on pin A0, prints the result to the serial monitor.

Digital signal on pin 10 decides to turn the LED on or off according to the analog input

Connect all input to the Arduino.

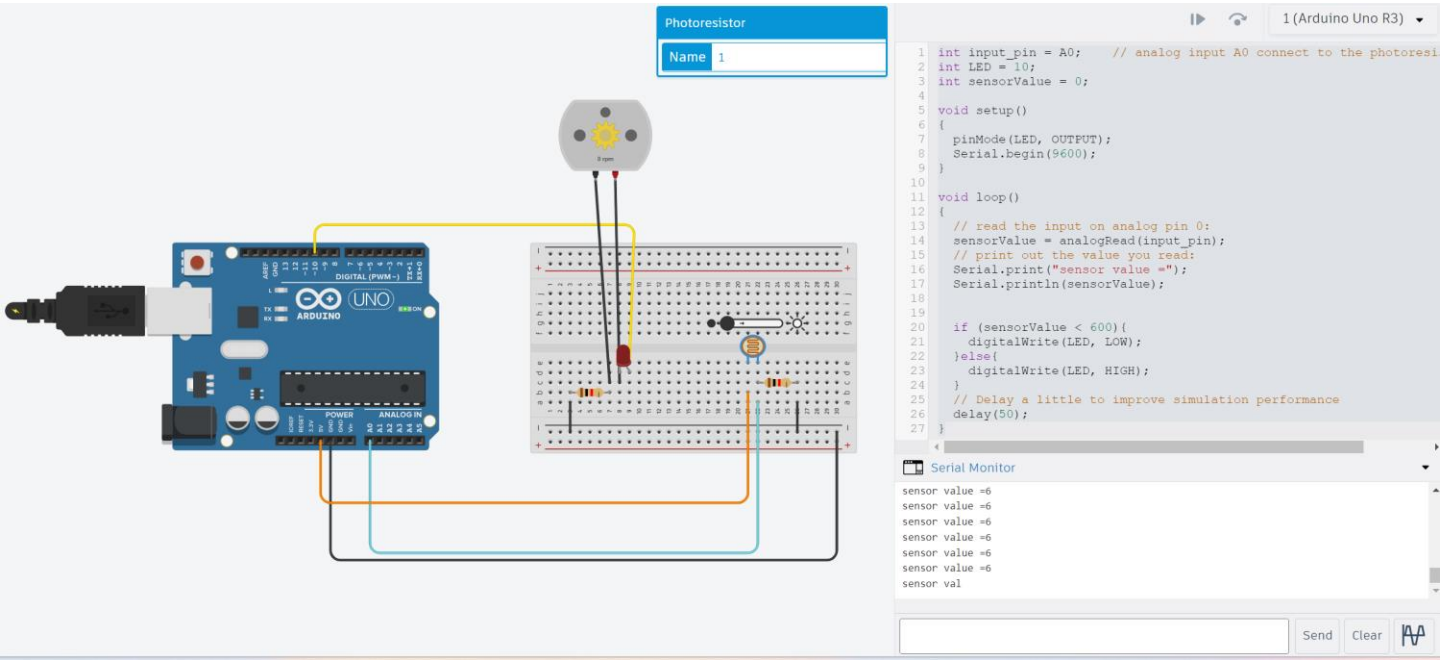
Source Code

```
int input_pin = A0;  // analog input A0 connect to the photoresistor sensor

int LED = 10;

int sensorValue = 0;

void setup()

{

  pinMode(LED, OUTPUT);

  Serial.begin(9600);

}

void loop()

{

  // read the input on analog pin 0:

  sensorValue = analogRead(input_pin);

  // print out the value you read:

  Serial.print("sensor value =");

  Serial.println(sensorValue);

 // control the motor and the LED

  if (sensorValue < 600) {

    digitalWrite(LED, LOW);

  } else {

    digitalWrite(LED, HIGH);

  }

   delay(50);     // Delay a little to improve simulation performance

}
```

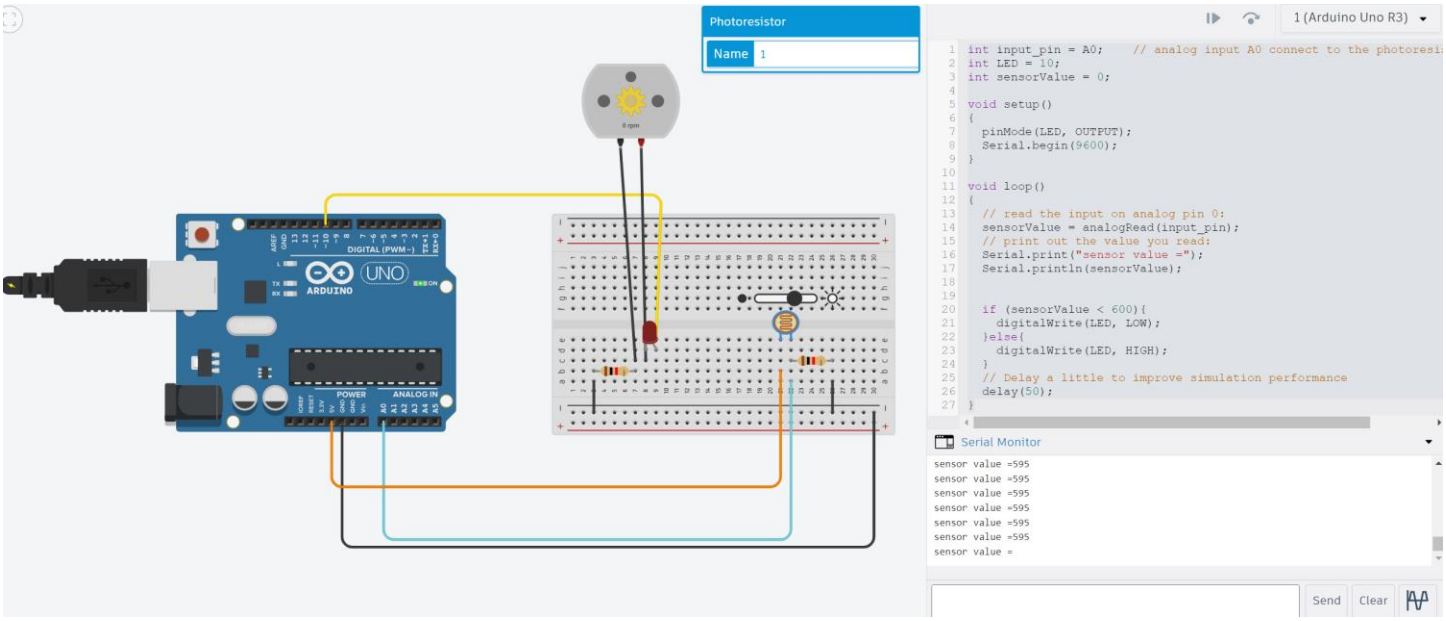
Result

Open serial monitor to view the output from the photoresistor

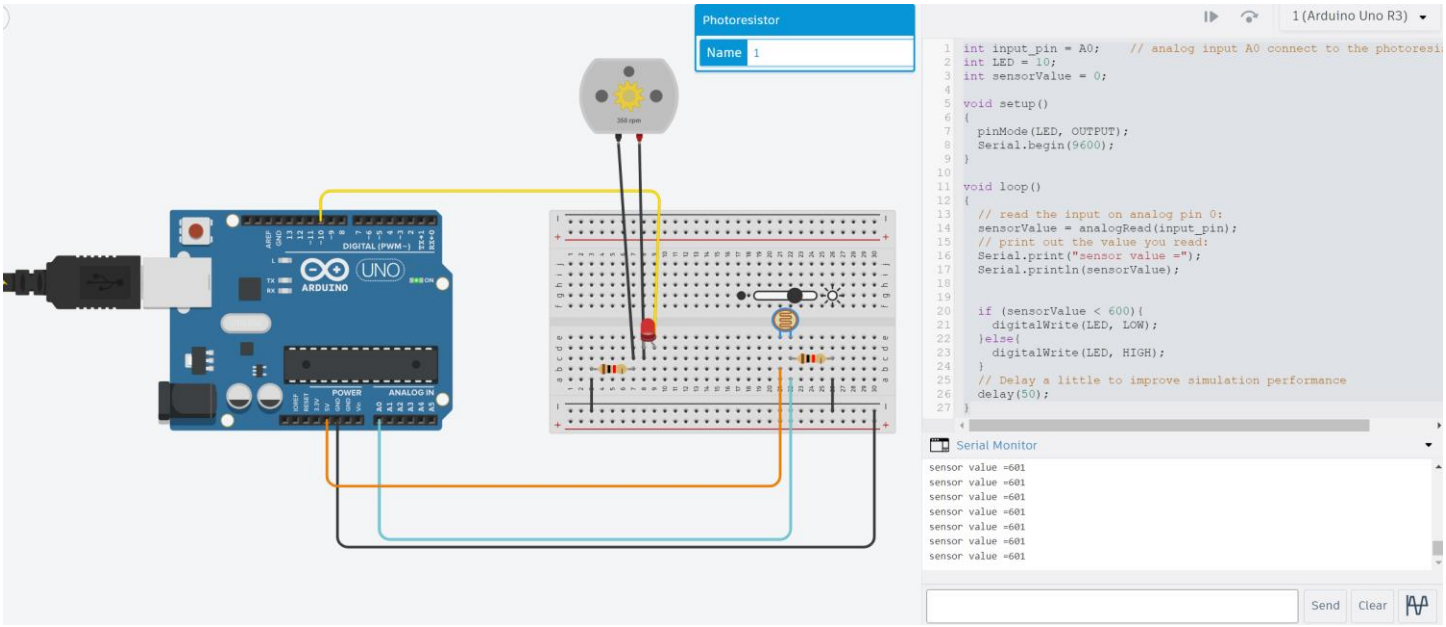
Low light intensity:



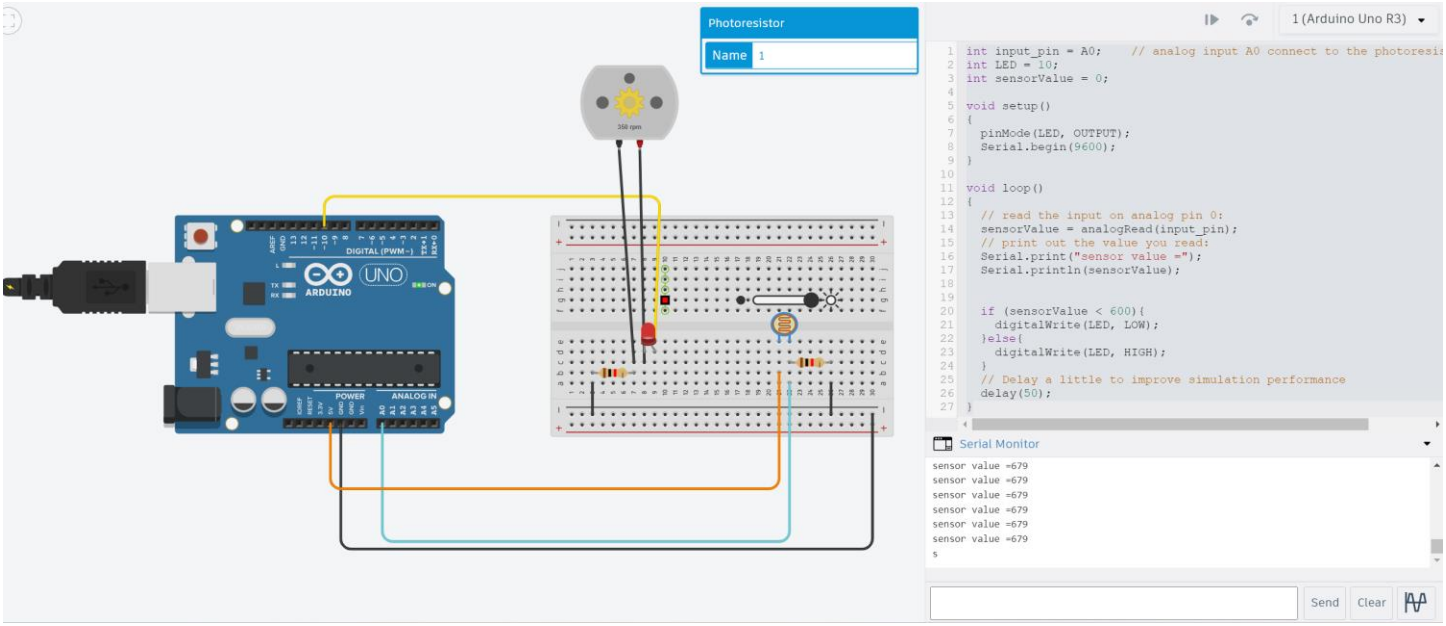
Light intensity = 595:



Light intensity = 601:

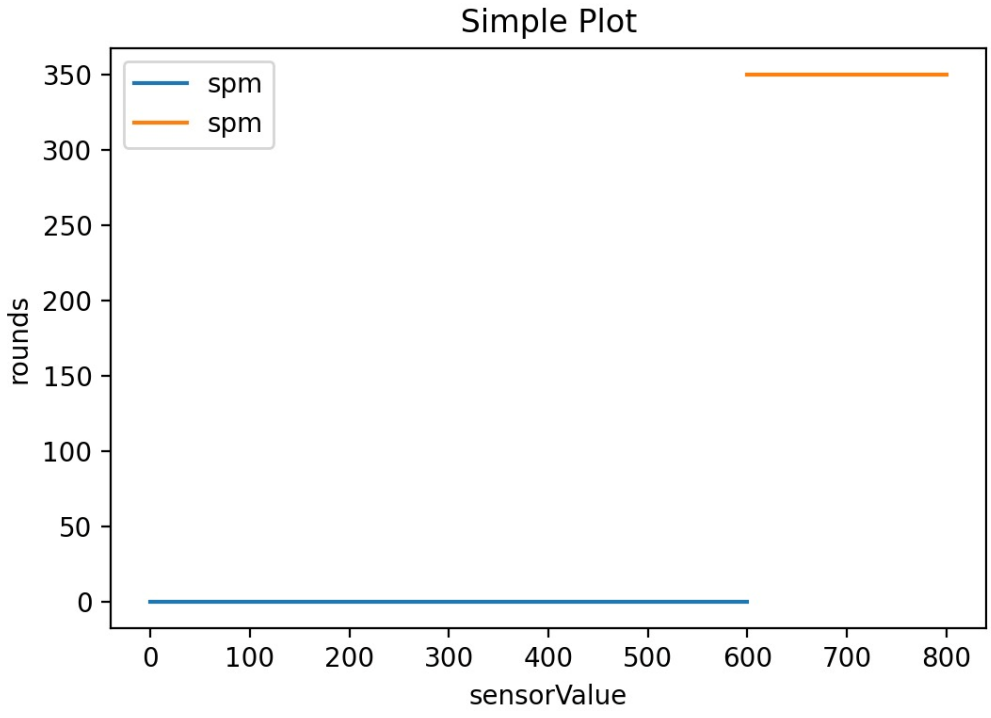


High light intensity:



Python with Arduino

```
import matplotlib.pyplot as plt

import numpy as np



x = np.linspace(0, 600, 10)

y = np.linspace(600,800,10)

fig, ax = plt.subplots(figsize= (6, 4), #layout='constrained')

ax.plot(x, x*0, label='spm') # Plot some data on the axes.

ax.plot(y, y*0+350, label='spm')

ax.set_xlabel('sensorValue')

ax.set_ylabel('rounds')

ax.set_title(" Simple Plot ")

ax.legend()

plt.show()
```



Contribution of group members

Wai Huen, Sin (Matriculation number: 30007257)	Yiheng, Cao (Matriculation number: 30007197)
- Arduino - Lab Report	- Python - Lab Report